

Analyzing the interface properties of cubic GaN/AlN heterostructures using impedance spectroscopy

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ABSTRACT

Cubic gallium nitride (c-GaN) is an attractive material system for various electronic and optoelectronic device applications, as it lacks internal piezoelectric polarization fields unlike its hexagonal counterpart. Yet, the fabrication of high-performing device structures depends on the growth of high-quality c-GaN thin films, which remains a challenging task. Here, we present a systematic characterization of the electrical properties of cubic GaN/AlN heterostructures with varying thicknesses grown by molecular beam epitaxy on 3C-SiC/Si templates. Impedance spectroscopy is used as a powerful tool for characterizing transport through the sample structure as well as the defect density at the existing interfaces. While the bare template exhibits a single feature in the impedance spectrum associated with the 3C-SiC/Si interface, a second low-frequency process arises under forward bias in samples with cubic GaN/AlN layers on top. Quantitative analysis of the impedance response reveals a reduction in the impurity density with increasing c-GaN film thickness. This improvement in structural quality with extended growth time is accompanied by a decrease in defect density at the interfaces. These results establish a direct link between optimized growth conditions and structural quality in cubic nitride heterostructures, providing a pathway toward the reliable design of next-generation electronic and optoelectronic devices.

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INTRODUCTION

The growing demand for high-performance semiconductor technologies across high-power electronics, radio-frequency systems, and optoelectronic applications necessitates both the improvement of established device architectures and the exploration of novel material platforms. Gallium nitride (GaN) has emerged as a key material for next-generation high-power applications owing to its wide bandgap and high breakdown fields.^{1–3} However, the thermodynamically stable wurtzite phase of GaN represents a fundamental limitation regarding the design of efficient optoelectronic device structures. Strong internal polarization fields, which arise at interfaces along the c-direction due to the hexagonal crystal structure, result in a reduction of the radiative recombination rate in quantum-confined systems.^{4–7}

Cubic GaN (c-GaN) as a metastable polymorph is free of internal piezoelectric fields and, thus, offers a pathway for the realization of improved GaN-based optoelectronic device structures, such as micro-light-emitting diodes or optical sensors.^{8–13} However, the epitaxial growth of high structural quality c-GaN thin films remains challenging, primarily due to the lack of native bulk c-GaN crystals or alternative lattice-matched, high-quality substrates with a zinc-blende crystal structure.¹² Heteroepitaxial growth on Si and GaAs templates is hindered by the relatively large lattice mismatch with c-GaN,^{14–17} which promotes defect formation at the interface and degrades crystalline quality. In contrast, 3C-SiC exhibits a much closer lattice match, but significant discrepancies between thermal expansion coefficients with c-GaN exist and induce thermal strain during cooling from typical growth temperatures, resulting in crack formation and compromised structural integrity.^{18,19}

As recently shown by Zscherp *et al.*, the structural quality of the c-GaN thin films grown by plasma-assisted molecular beam epitaxy (PAMBE) on 3C-SiC/Si templates can be improved significantly by suitable pre-growth treatments and the introduction of an additional c-AlN buffer layer between the template and the c-GaN layer.²⁰ In addition, this provides a good template for the subsequent growth of c-(In,Ga)N over the entire composition range.^{21–24} This offers the possibility for the realization of (In,Ga)N/GaN quantum well structures for optical devices,²⁵ such as light-emitting diodes or lasers with wavelength covering the whole visible spectrum from near-ultraviolet to the near-infrared.^{13,21,22,25,26}

Interfaces play a crucial role in the functionality of all electronic and optoelectronic devices.^{27–30} In particular, interfaces of high structural quality and low defect density are essential for achieving high device performances, making a comprehensive characterization of their properties necessary. Among the most widely employed techniques for probing interface properties of semiconductor heterostructures are capacitance–voltage (C–V) measurements or impedance spectroscopy (IS).^{31–37} These methods not only allow the characterization of interface trap states but also give valuable insights into transport processes across the interface as well as the capacitance of the device structure. In this work, we use impedance spectroscopy for characterizing the interface properties of c-GaN/c-AlN heterostructures, which were grown on 3C-SiC/Si templates using PAMBE. A detailed characterization of the impedance of heterostructures with varying c-GaN layer thicknesses reveals that the density of interface states decreases with increasing layer thickness. This reduction correlates with an increased structural quality of the thin films. One possible reason for the increased structural quality is the extended growth time of the thicker c-GaN layers, which allows defects to heal more effectively. These findings confirm the improved structural quality of the c-GaN thin films grown on c-AlN buffer layers and underscore the critical importance of carefully optimized growth conditions for the deposition of quantum structures of high structural integrity.

EXPERIMENTAL

The impedance of five different sample structures is analyzed in this work. The first sample consists of the 3C-SiC/Si template only, while the second sample has an additional 8 nm thin c-AlN buffer layer on top of the 3C-SiC/Si template. Samples three to five consist of the full c-GaN/c-AlN/3C-SiC/Si sample stack with an 8 nm thin c-AlN layer and a thickness of the c-GaN layer of 200, 400, and 1800 nm, respectively. In the following, these samples are labelled according to their surface layer thickness. The 3C-SiC/Si substrates were purchased from NovaSiC. All samples are rectangular in shape and their cross-sectional areas are given in Table S2 in the [supplementary material](#). The 3C-SiC layer and the Si substrate are n-type with a target doping concentration of about $N_D = 5 \times 10^{17} \text{ cm}^{-3}$ and thickness of 10 and 525 μm , respectively. The Si surface has a (100) orientation. The c-AlN buffer layer and the c-GaN layers were grown by PAMBE as described in detail by Zscherp *et al.*²⁰ After a short aluminum flash procedure for surface cleaning at 850 °C, the sample temperature was lowered to 720 °C for the growth process. The growth of the layers was performed under slightly metal-rich conditions. The growth time of

the AlN layer was 1 min. The growth time of the c-GaN layer was increased from 37 to 60 and 300 min for the growth of 200, 400, and 1800 nm thick layers, respectively. As shown in Fig. S1 in the [supplementary material](#), x-ray diffraction measurements confirm the cubic structure.²¹ Scanning transmission electron microscopy (STEM) analysis confirms the presence of stacking faults within the c-GaN layers, which partly annihilate within the first few hundred nanometers of c-GaN.²⁰

For the characterization of interface properties, impedance spectroscopy (IS) measurements were performed. The impedance of the heterostructures was measured in a two-electrode setup with a Zennium Pro potentiostat (Zahner-Elektrik) inside a Novocool Cryosystem from Novocontrol Technologies. Electrical contacts were realized by two brass electrodes pressed against the top side and the bottom side (Si substrate) of the samples. The impedance measurements were conducted using an AC frequency ranging from 0.1 Hz to 8 MHz and an AC amplitude of 5 mV. The bias voltage V_{DC} was varied in steps of 0.5 V in the range between -4 V and $+4 \text{ V}$ with respect to the silicon layer (ground level) at six different temperatures between -100 and $+100$ °C.

RESULTS AND DISCUSSION

Figures 1(a)–1(f) show the measured impedance of the 3C-SiC/Si template, the template with the 8 nm c-AlN layer only, and the sample stack with the 200 nm c-GaN layer, respectively. All spectra are plotted in Nyquist representation, where the negative imaginary part of the impedance $-Z_{\text{Im}}$ is plotted vs its real part Z_{Re} . The Nyquist plot for the impedance of the 3C-SiC/Si template shows only one semicircle [Figs. 1(a)–1(b)] with an offset from the origin of about $R_0 = 50 \text{ }\Omega$ for all applied voltages. The semicircle increases in diameter with increasing DC bias voltages, reaching the largest value at $V_{\text{DC}} = 0 \text{ V}$, and decreases again at higher positive voltage [Fig. 1(b)]. For the sample with the c-AlN layer on top [Figs. 1(c)–1(d)], a comparable impedance spectrum is observed for negative DC bias voltages (reverse bias), including an offset resistance and only one semicircle. The finite resistance at low frequencies clearly demonstrates that transport through the c-AlN layer occurs despite its large bandgap, probably due to defects present in the layer. However, at positive DC bias voltages (forward bias) [Fig. 1(d)], a second semicircle appears in the frequency range below 10^5 Hz . Again, the finite impedance at low frequencies clearly shows that electrical transport occurs through the sample structure, i.e., the AlN thin film. A comparable behavior is observed for the sample with GaN/AlN heterointerface [Figs. 1(e)–1(f)], indicating a negligible impact of the 200 nm thin GaN layer on the total impedance. The same holds for the samples with 400 and 1800 nm thick c-GaN layers, whose impedance response is shown in Figs. S2(g)–S2(j) in the [supplementary material](#). All samples exhibit comparable rectifying current–voltage characteristics (see Fig. S3 in the [supplementary material](#)), revealing the strong impact of the heterointerfaces on the overall electrical properties.

For the qualitative analysis of the measured impedance and the extraction of the relevant transport parameters, the impedance can be described by an equivalent circuit consisting of resistance R_0 connected in series with two RQ-elements^{38,39} as depicted in Fig. 1(g).

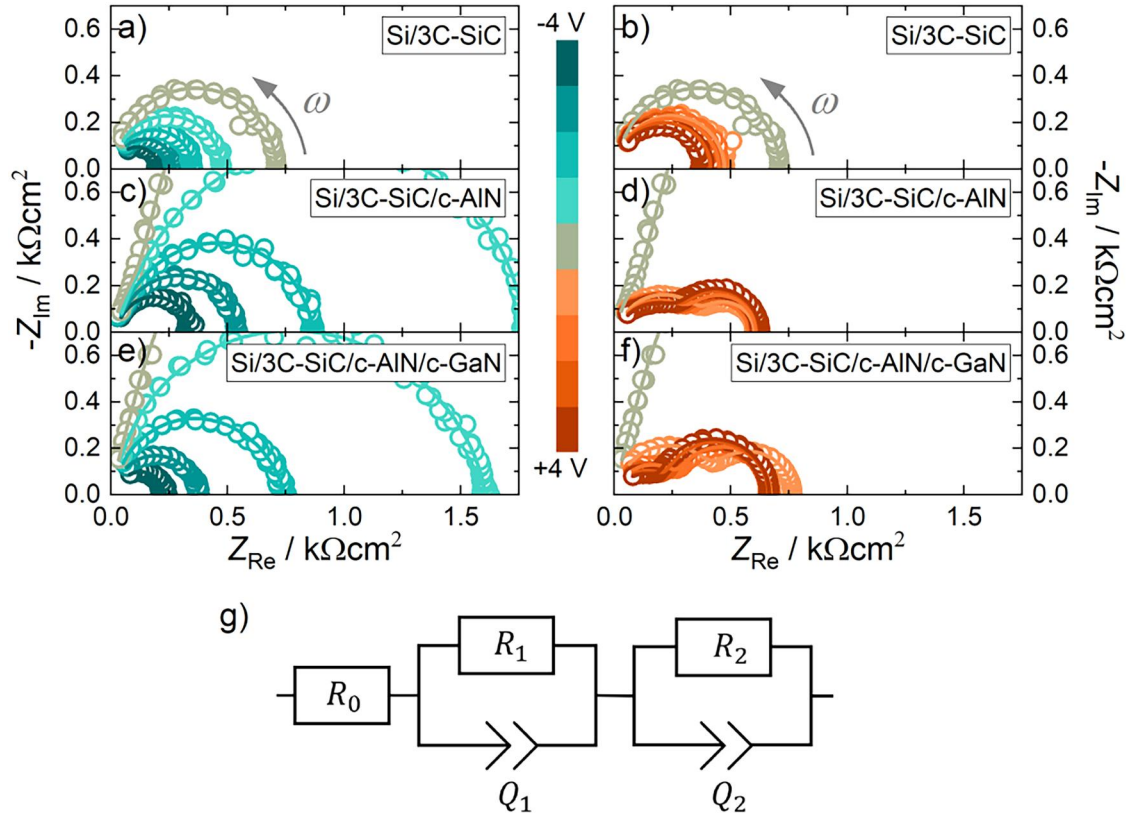


FIG. 1. Area-normalized IS data (circles) of three different samples and their corresponding equivalent circuit fit results (solid lines) at negative (left) and positive (right) DC voltages at $T = 0^\circ\text{C}$. (a) and (b) 3C-SiC/Si template alone, (c) and (d) template with 8 nm c-AlN, and (e) and (f) template with 8 nm c-AlN and 200 nm c-GaN. (g) Equivalent circuit used for the fitting procedure.

The resistance R_0 represents the sum of the bulk resistances of the different layers, as well as the internal resistance of the measurement setup resulting from the measurement cables, their connections, and the electrode contacts.³¹ Each of the two RQ-elements represents one of the observed semicircles and describes a transport process within the sample structure. We chose constant phase elements (CPE) Q_i instead of capacitances C_i to account for minor non-idealities of the semicircles.³¹ However, the related capacitances can be calculated from the CPE parameters α_i and Q_i according to^{40–42}

$$C_i = (R_i^{1-\alpha_i} Q_i)^{1/\alpha_i}. \quad (1)$$

The parameter α_i describes the non-ideality of the capacitance with $\alpha_i = 1$ corresponding to an ideal capacitive behavior. The characteristic relaxation time of the transport process are then given by^{43,44}

$$\tau_i = (R_i Q_i)^{1/\alpha_i} = R_i C_i. \quad (2)$$

As all samples show at least one semicircle, the first transport process is ascribed to the 3C-SiC/Si template. The bulk resistances

of the Si and 3C-SiC layers can be estimated using the following relation:

$$R_i = \rho_i \cdot \frac{d_i}{A}, \quad (3)$$

where ρ_i is the resistivity, A is the sample area, and d_i is the layer thickness. Using the values $\rho_{\text{Si}} = 5 \text{ }\Omega\text{cm}$ taken from the NovaSiC datasheet and $\rho_{\text{3C-SiC}} = 1 \text{ }\Omega\text{cm}$,⁴⁵ the resistances of the 3C-SiC layer and the Si substrate are estimated to be in the order of 10^{-1} and $10^{-5} \text{ }\Omega$, respectively. These values are significantly smaller than the DC resistance of the sample as given by the real part of the impedance at low frequencies. An estimation of the relaxation time associated with charge transport through the Si substrate or the SiC layer (see Table S1 in the [supplementary material](#)) yields characteristic frequencies of the order of $\omega = \tau^{-1} \approx 10^{14} \text{ Hz}$, which lie well above the probed range. Consequently, the first semicircle, and the corresponding parameters (R_1 and C_1), can be ascribed to the 3C-SiC/Si interface and non-ohmic contact resistance contributions rather than a bulk process.^{39,46} Thus, the resistance of the RQ element reflects charge transport through the interface, while the related capacitance represents the charge separation at the interface. Following the same considerations as before, the

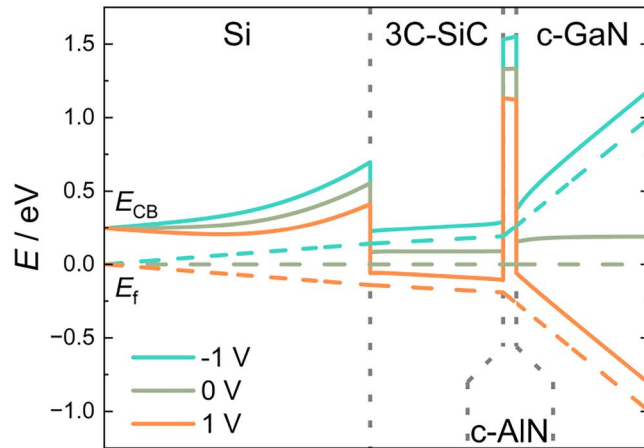


FIG. 2. Schematic depiction of the conduction band edge (solid lines) and the quasi-Fermi level (dashed lines) at different DC bias based on qualitative simulations with nextnano++.

resistance contribution of the additional 8 nm thin c-AlN layer can be estimated to be of the order of $R_{c\text{-AlN}} \approx 10^{-3} \Omega$ ($\rho_{\text{AlN}} \approx 3 \times 10^2 \Omega\text{cm}$)⁴⁷ with a characteristic frequency of the order of $\omega_{c\text{-AlN}} \approx 10^{10}$ Hz (see Table S1 in the [supplementary material](#)). Thus, the second semicircle arising at frequencies below 10^5 Hz and at positive DC voltages in the impedance of the c-AlN sample is attributed to the c-AlN/3C-SiC interface. Interestingly, the introduction of an additional c-GaN layer and the related c-GaN/c-AlN interface has no qualitative impact on the measured impedance. The second process and its corresponding parameters (R_2 and C_2), therefore, represent the transport through both the 3C-SiC/c-AlN interface and the c-AlN/c-GaN interface. It is worth noting that the area-normalized diameter of the first semicircle varies among the samples, which is attributed to differences in the contact resistances, demonstrating its impact on the first process. Based on these variations, the contact resistance is estimated to be in the order of a few k Ω . However, in the following, the discussion focuses on the properties of the 3C-SiC/c-AlN and the c-AlN/c-GaN interfaces, which are associated with the second semicircle and are, therefore, not directly affected by variations in the contact resistance.

In the following, we will refer to the transport across the 3C-SiC/Si and the c-AlN/3C-SiC (and c-GaN/c-AlN) interfaces as the first and second processes, respectively. These findings are supported by qualitative, bias-dependent simulation of the band structure using nextnano++. A schematic depiction of the conduction band edge with applied bias is presented in Fig. 2. As shown in Fig. S4(a) in the [supplementary material](#), the band structure relative to the quasi-Fermi level at the 3C-SiC/Si interface remains nearly independent of the applied bias voltage V_{DC} . This agrees with our experimental findings, where only minor changes of the diameter of the first semicircle are observed for varying V_{DC} . In addition, the qualitative simulations in Fig. 2 [and Figs. S4(b) and S4(c) in the [supplementary material](#)] show that the relative band bending at the c-AlN/3C-SiC interface and the c-GaN/c-AlN

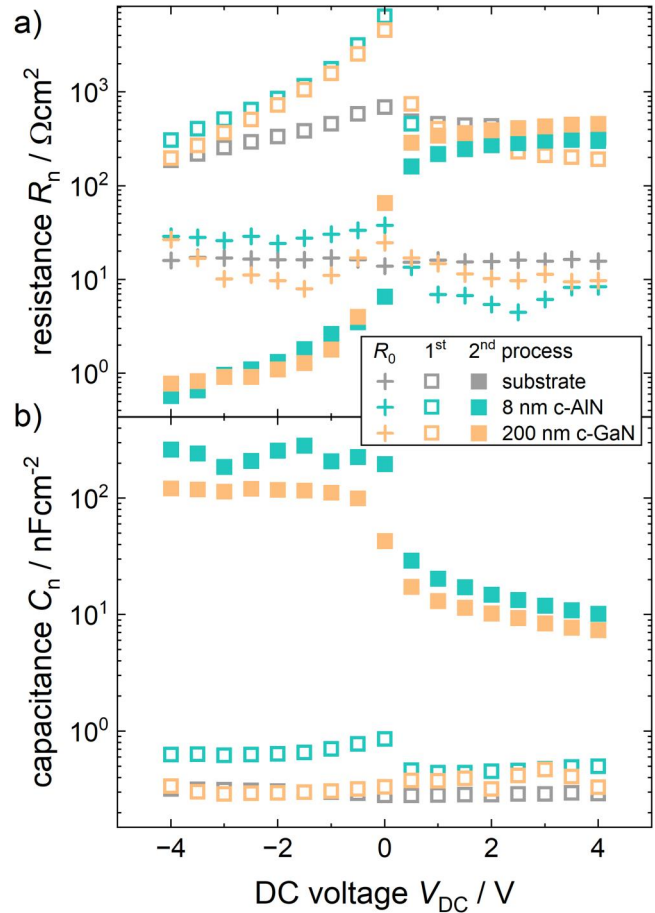


FIG. 3. Area-normalized fit parameters of the equivalent circuits vs DC voltage for three different samples at 0 °C. (a) resistance; (b) capacitance. Empty and filled squares correspond to the first and second processes, respectively, and crosses represent the series resistance.

interface exhibits a strong bias dependence. This explains the pronounced bias dependence of the second process, which is ascribed to the transport through these interfaces.

For fitting of the impedance, we used so-called ZARC elements instead of the RQ-elements. A ZARC element only differs from an RQ-element by its fitting parameters, as it yields resistance and relaxation time as fit parameters instead of the resistance and capacitance values. The fit parameters of a ZARC and an RQ element can be converted from one to the other using Eq. 2. This allows for a better assignment of the transport process to the corresponding time constant. The results of the fitting are shown in Fig. 1 (and Fig. S2 in the [supplementary material](#)) as solid lines, revealing that the equivalent circuit can precisely match the measured data. Figures 3(a) and 3(b) show the determined area-normalized resistances $R_{n,i}$ and capacitance $C_{n,i}$ values, respectively, for the first process (empty squares) and the second process (filled squares). Area-normalized values are plotted in

Fig. 3 for better comparison, as the samples deviate slightly in size. The complete set of fit parameters and the corresponding time constants of all samples are presented in Fig. S5 in the [supplementary material](#). For all samples, the fit of the impedance yields small values for the bulk and measurement setup resistances R_0 ranging between 10 and 90 Ω .

For all samples, the resistance and capacitance values associated with the 3C-SiC/Si interface (first process) exhibit consistent qualitative behavior. The values of the fit parameter α are shown in Fig. S6 in the [supplementary material](#). Only minor deviations from the ideal capacitive behavior ($\alpha = 1$) are observed, which are attributed to slight inhomogeneities at the heterointerfaces. The capacitance remains essentially independent of applied bias voltage, whereas the resistance shows a pronounced asymmetric increase as the DC voltage approaches 0 V from both positive and negative directions. This voltage dependence of the resistance is characteristic for a 3C-SiC/Si heterojunction.^{48–50} In contrast, a distinctly different voltage dependence is observed for the parameters of the second process, which influences the impedance spectra only under forward bias for the samples incorporating a c-AlN layer. At negative DC voltage, the resistance $R_{n,2}$ of the second process is at least two orders of magnitude lower than $R_{n,1}$, increases slightly, and finally shows a step-like increase by two orders of magnitude when approaching 0 V. For positive DC voltage, the resistance $R_{n,2}$ slightly increases further reaching comparable values as $R_{n,1}$. The capacitance $C_{n,2}$ in the negative voltage range is nearly constant and at least two orders of magnitude higher than $C_{n,1}$. At 0 V, a drop occurs, followed by a further decrease in $C_{n,2}$ with increasing bias voltage. It is worth noting that the fit parameters of the second process can only be estimated, as the second process cannot be resolved in the impedance spectrum. Thus, the values for the parameters $R_{n,2}$ and $C_{n,2}$ under negative bias only represent upper and lower limits, respectively. Otherwise, the second transport process would be visible in the impedance spectrum also under reverse bias. However, the voltage dependence of $C_{n,2}$ is indicative for the presence of a space-charge region at the c-GaN/c-AlN/3C-SiC interfaces. Variations in the thickness of the c-GaN layer on top only result in minor changes in resistance and capacitance values associated with the second process. Qualitative band diagram simulations of the heterointerfaces [Fig. S4(c) in the [supplementary material](#)] reveal that a space-charge region is also present within the c-GaN layer at the c-GaN/c-AlN interface. Notably, the band bending in the c-GaN layer exceeds that on the 3C-SiC side. However, the parameters of the second process comprise the properties of both the c-GaN/c-AlN and the c-AlN/3C-SiC interface. Based on the capacitance values of the second process, the properties of the space-charge region can be estimated following the Mott-Schottky approach:^{51–54}

$$\frac{1}{C_{n,2}^2} = \frac{A^2}{C_2^2} = \frac{2}{\epsilon\epsilon_0\epsilon_r N_D} \left(V_i - V_{fb} - \frac{k_B T}{e} \right), \quad (4)$$

where e is the elementary charge, ϵ_0 is the vacuum permittivity, ϵ_r is the permittivity of the semiconductor, A is the area of the semiconductor, k_B is the Boltzmann constant, and T is the absolute temperature. N_D denotes the impurity (dopant) density of the

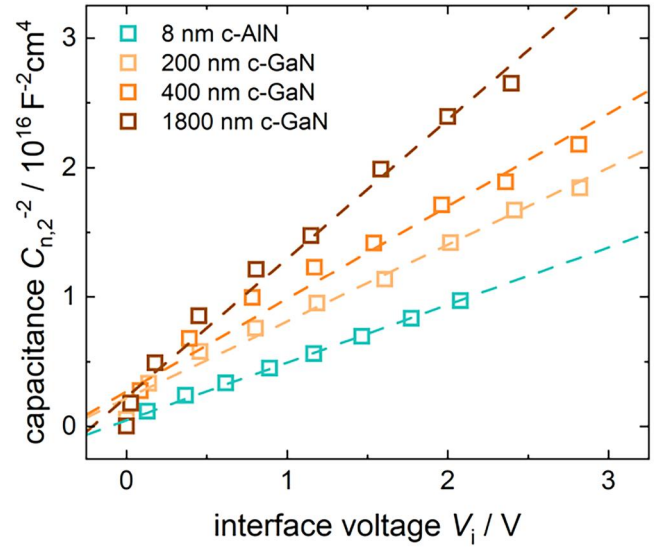


FIG. 4. Mott-Schottky plot of the area-normalized interface capacitance for different samples with c-AlN/3C-SiC and c-GaN/c-AlN interfaces at $T = 0^\circ\text{C}$. The dashed lines represent linear fits to the experimental data.

semiconductor. The so-called flatband voltage V_{fb} describes the DC voltage at which the bands are flat. V_i denotes the fraction of the applied DC bias that drops across the nitride-based heterointerfaces, i.e., across the resistance R_2 in the equivalent circuit shown in Fig. 1(g). This voltage drop can be calculated from the applied DC bias V_{DC} according to

$$V_i = V_{DC} \cdot \frac{R_{n,2}}{R_{n,0} + R_{n,1} + R_{n,2}}. \quad (5)$$

For heterostructures exhibiting unsymmetric band bending, the Mott-Schottky analysis provides, to first approximation, the impurity density of the semiconductor with lower impurity concentration.⁵⁵ Given that the band bending within the c-GaN layer is more pronounced than at other interfaces within the structure (see Fig. S4 in the [supplementary material](#)), the impurity density extracted from the analysis is attributed primarily to the c-GaN layer of the sample structures investigated.

As shown in Fig. 4, $C_{n,2}^{-2}$ exhibits a nearly ideal, linear behavior at positive voltages (forward bias) for all samples investigated. The slight deviations from the linear behavior at low interface voltages can be attributed to interface states.^{56,57} The positive slope is typical for n-type semiconductors and indicates the presence of donor impurities in the nominal undoped cubic nitride layers. The flatband potential V_{fb} is about (-0.2 ± 0.2) V and, therefore, in good agreement with the qualitative band simulations in Fig. S4 in the [supplementary material](#), which confirm flatbands at a DC voltage of about 0 V.

Figure 5 shows the impurity density as a function of c-GaN thickness, extracted from the slopes of the linear fits in Fig. 4, with a relative error below 5%. For the sample with a single c-AlN layer

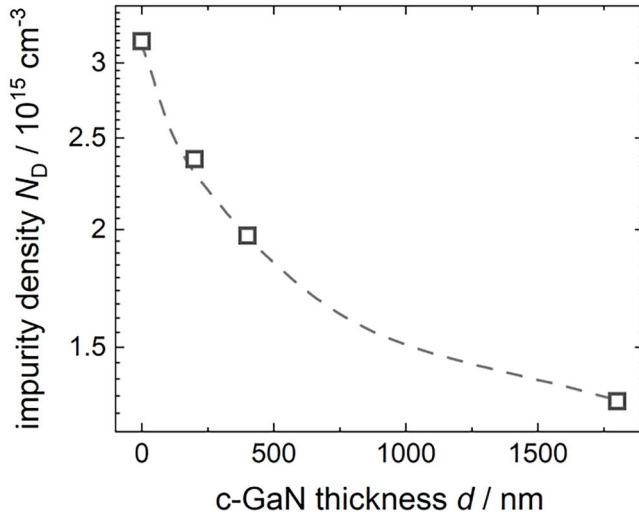


FIG. 5. Derived impurity density plotted against the thickness of the c-GaN layer. Note that a c-GaN thickness of 0 nm corresponds to the c-AlN sample. The dashed line is a guide for the eye.

only, an impurity density of about $3.1 \times 10^{15} \text{ cm}^{-3}$ is obtained. This impurity density is attributed to electrically active defects in the non-intentionally doped AlN layer, whose concentration can be assumed to be significantly lower compared to the dopant concentration of the SiC layer.⁵⁵ With increasing c-GaN thickness, a decrease in the impurity density is observed, reaching values of about $1.3 \times 10^{15} \text{ cm}^{-3}$ for the sample with a 1800 nm thick c-GaN layer. This decrease indicates an improved crystal quality, which can be correlated with the longer growth time of the thicker films, promoting defect reduction. In particular, the annihilation of stacking faults within the initial c-GaN layers has been reported,²⁰ leading to a lower defect density in the subsequently grown material. As the depletion width within the c-GaN layer exceeds 100 nm, as estimated by the qualitative nextnano++ simulations [Fig. 2; Fig. S4(c) in the [supplementary material](#)], thicker c-GaN layers exhibit a lower average stacking fault density and, thus, a lower defect density consistent with the trend observed in Fig. 5.

To gain a better understanding of the interface properties and the electrical transport through the sample structure, the capacitance-voltage (C-V) characteristics as well as the frequency-dependent parallel conductance G_p were analyzed. The overall parallel capacitance C_p and the parallel conductance G_p are calculated from the measured impedance,⁵⁸

$$C_p = \frac{-Z_{\text{Im}}}{2\pi f \cdot (Z_{\text{Re}}^2 + Z_{\text{Im}}^2)}, \quad (6)$$

$$G_p = \frac{Z_{\text{Re}}}{Z_{\text{Re}}^2 + Z_{\text{Im}}^2}, \quad (7)$$

where Z_{Re} and Z_{Im} are the real and imaginary parts of the impedance and f is the applied AC frequency.

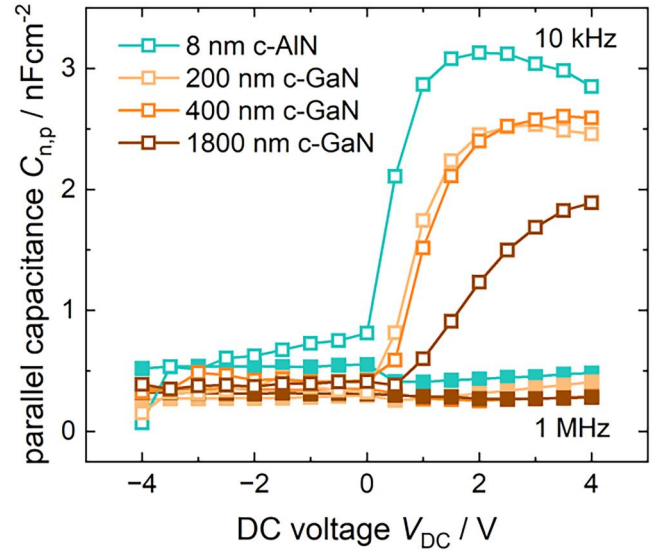


FIG. 6. Calculated C-V characteristics of the area-normalized parallel capacitance at high (1 MHz, filled symbols) and low frequencies (10 kHz, open symbols) and $T = 0^\circ \text{C}$ for the sample structures investigated.

Figure 6 shows the C-V characteristics of the area-normalized parallel capacitance $C_{n,p}$ of the sample structures investigated at high and low frequencies of 1 MHz and 10 kHz, respectively. At high frequencies, i.e., for $f > 10^5 \text{ Hz}$, the parallel capacitance remains constant for reverse bias and drops to only slightly lower values under forward bias. The value of $C_{n,p} \approx 0.2 \text{ nF}$ corresponds to the capacitance value $C_{n,1}$ of the first process [see Fig. 3(b)] and, thus, arises from the band bending at the 3C-SiC/Si interface as shown in Fig. 2. At these high frequencies, the frequency change is too fast for the accumulation of electrons at the c-AlN/3C-SiC interface. Thus, the total capacitance is dominated by $C_{n,1}$ only, as also observed in the impedance. From the values of $C_{n,p}$, the width of the depletion region W_D at the 3C-SiC/Si interface can be estimated ($\epsilon_r^{\text{SiC}} \approx 10$),^{59–61}

$$C_{n,1} = \frac{C_{\text{SiC/Si}}}{A} = \epsilon_0 \epsilon_r \frac{1}{W_D} \approx 0.25 \text{ nFcm}^{-2} \Rightarrow W_D \approx 35 \mu\text{m}. \quad (8)$$

Thus, the band bending at the 3C-SiC/Si interface widely extends into the Si substrate by several micrometers independent of the applied voltage, in agreement with our qualitative simulations [Fig. S4(a) in the [supplementary material](#)].

At frequencies below 10^5 Hz , the impedance response is dominated by the second process. Thus, the parallel capacitance $C_{n,p}$ at low frequencies reflects the capacitance $C_{n,2}$ describing the combined properties of the c-AlN/3C-SiC and the c-GaN/c-AlN interface. $C_{n,p}$ shows a step-like behavior, which is characteristic for heterostructures or metal-insulator structures (MIS).³⁴ At negative V_{DC} (reverse bias), the parallel capacitance reaches values comparable to the depletion capacitance $C_{n,1}$ of the 3C-SiC/Si

interface, as no charge accumulation occurs at the c-AlN/3C-SiC interface. However, a steep increase occurs at around 0 V, and $C_{n,p}$ reaches a value of about 2.5 nFcm^{-2} for the sample with the c-AlN layer. This increase in parallel capacitance arises from the accumulation of charge carriers at the c-AlN/3C-SiC interface and corresponds with the depletion capacitance $C_{n,2}$ of the c-AlN/3C-SiC interface [see Fig. 3(b)]. After the step-like increase, the parallel capacitance slightly decreases again under forward bias. This behavior is well known to be caused by the presence of a series resistance, which accounts for the non-insulating properties of the AlN layer.⁶² The C-V characteristics of the samples with an additional c-GaN layer shown in Fig. 6 exhibit only slight deviations from the c-AlN sample. With increasing c-GaN layer thickness, the increase in $C_{n,p}$ gets less steep, underlining the impact of the c-GaN layer on the measured impedance. Qualitative simulations of the band structure (Fig. S4 in the [supplementary material](#)) confirm the changes in band bending with applied bias. At the c-AlN/3C-SiC interface, the 3C-SiC conduction band bends upward for negative bias, leading to an electron depletion. Under forward bias, a downward band bending arises, leading to an accumulation of electrons. The bands of the c-AlN layer [Fig. S4(b) in the [supplementary material](#)] are only tilted under bias, i.e., no bending occurs, which fits its highly resistive properties.

To gain additional information about the interface states of the prepared structures, the conductance method was used, where the frequency dependence of the area-normalized parallel conductance $G_{n,p}$ is analyzed.^{63–65} The conductance method was first discussed in detail by E. H. Nicollian and A. Goetzberger in 1967 and has since become a widely accepted and validated technique for determining interface trap state densities.^{65–67} Figure 7 shows

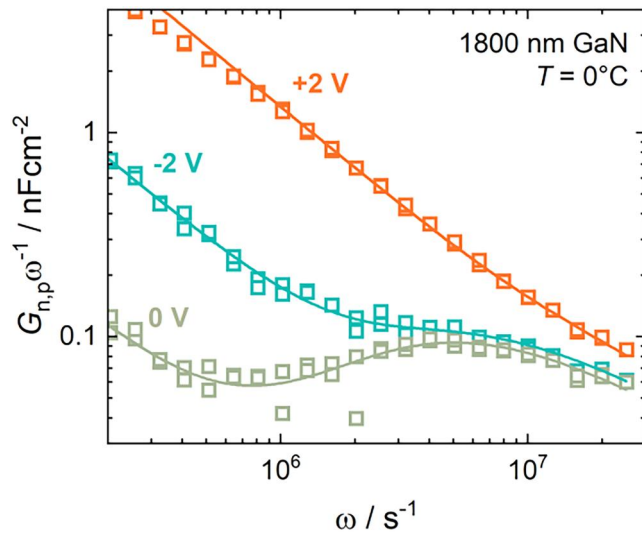


FIG. 7. Frequency dependence of the area-normalized parallel conductance of the 1800 nm sample at $T = 0^\circ\text{C}$ divided by radial frequency. The solid lines represent the fit to the experimental data (squares) using the conductance model.

$G_{n,p}/\omega$ of the heterostructure with 1800 nm thick c-GaN layer for three different bias voltages as a function of frequency in a double logarithmic representation. The linear decrease at lower frequencies is characteristic for the free carrier contribution (Drude response) of the sample structure described by the total DC resistance $R_S = R_0 + R_1 + R_2$ of the sample structure at low frequencies ($\omega \rightarrow 0$).^{33,68} At higher frequencies above 1 MHz and at negative bias, an additional conductance contribution arises for all samples except the template sample. This additional contribution is characteristic for interface trap states.^{69–72} As it only occurs for the samples with nitride layers, the 3C-SiC/Si interface as the origin of this conductance contribution can be eliminated. The influence of resistance R_S , a continuous distribution of interface trap states with density D_{it} , and their response times τ_{it} on the parallel conductance can be quantified by^{33,35,37,70}

$$\frac{G_{n,p}}{\omega} = \frac{G_p}{\omega A} = \frac{1}{R_S \omega A} + \frac{q^2 D_{it}}{2\omega \tau_{it}} \cdot \ln(1 + \omega^2 \tau_{it}^2), \quad (9)$$

where the interface trap density D_{it} is defined as the derivative of the interface trap charge density with respect to energy and q is the charge of the interface trap states.⁷³ The corresponding fits are represented by solid lines in Fig. 7. The interface trap-related conductance contribution appears only under negative bias. This behavior is consistent with the electron accumulation at the AlN/3C-SiC interface under positive bias [see also qualitative band bending simulation in Fig. S4(b) in the [supplementary material](#)], which increases the overall conductance of the sample structure and effectively masks the interface trap response. In contrast, under negative bias, electron depletion reduces the overall conductance, allowing the interface trap states to become clearly visible.

Apart from the trap state density, their energy level with respect to the conduction band edge ($E_C - E_{it}$) can be determined from the response time τ_{it} according to Shockley-Read-Hall statistics,^{65,74,75}

$$\tau_{it} = \frac{1}{v_{th} \sigma_n N_c} \cdot \exp\left(\frac{E_C - E_{it}}{k_B T}\right), \quad (10)$$

where v_{th} is the thermal velocity, σ_n is the electron capture cross section, and N_c is the effective density of states in the conduction band. Figure 8 shows the calculated trap density for all heterostructures investigated using the values $v_{th} = 1.9 \times 10^7 \text{ cm/s}$, $\sigma_n = 10^{-14} \text{ cm}^2$, and $N_c = 1.23 \times 10^{19} \text{ cm}^{-3}$ reported for trap states at the AlN/4H-SiC interface.⁷² Among these parameters, v_{th} exhibits the highest uncertainty, with reported values ranging between 20% and 200% of the nominal value.⁷⁶ Consequently, the uncertainty in the extracted interface-trap energy level is estimated to be about 15%. Nevertheless, it is noteworthy that the obtained interface trap densities are lower than those typically reported for hexagonal GaN-based heterostructures or MIS devices, indicating an overall improvement in interface quality.^{34,35,72,77–79} The reduced interface trap density may be induced by three different factors. First, cubic surface orientations generally exhibit lower interface densities compared to hexagonal orientations, as demonstrated in literature particularly for silicon-based systems.^{80–83} Second, MBE

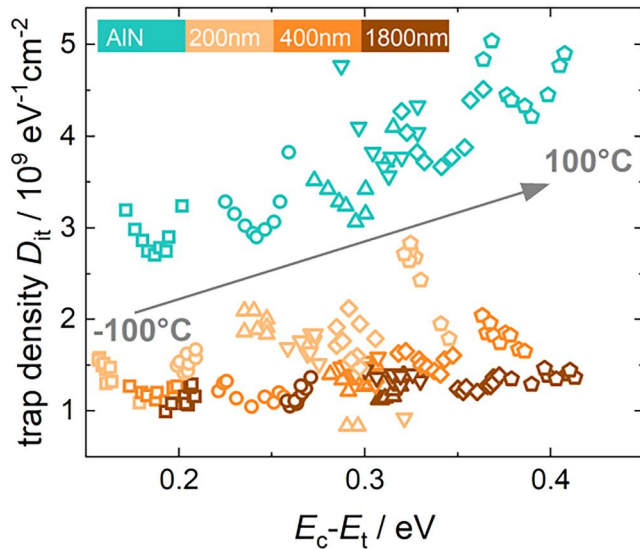


FIG. 8. Energy level distribution of the interface trap states at the c-AlN/3C-SiC interfaces for heterostructures investigated (color scale) and at different temperatures.

enables the growth of high-quality crystalline layers and abrupt interfaces, thereby reducing the interface trap density compared to other growth methods.^{63,64} Finally, the small lattice mismatch between the template and nitride layers suppresses the formation of interface defects and dislocations.^{84,85} Consequently, the

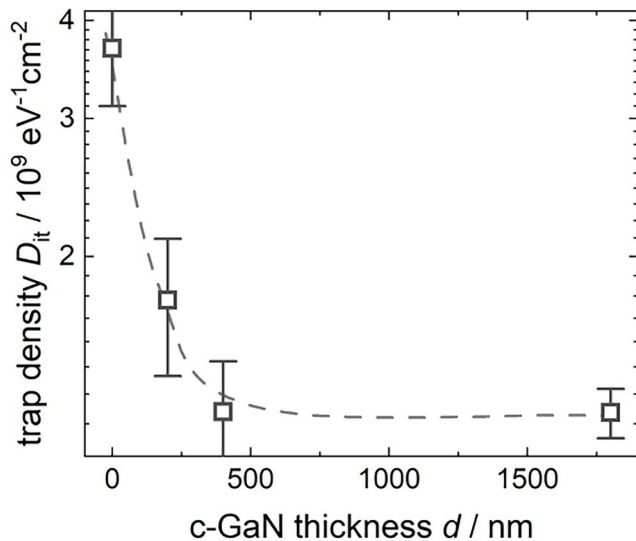


FIG. 9. Averaged interface trap density at the c-AlN/3C-SiC interfaces for varying c-GaN thickness and varying temperatures. Note that a c-GaN thickness of 0 nm corresponds to the c-AlN sample. The dashed line is a guide for the eye.

interface trap state density in cubic GaN heterostructures is at least two orders of magnitude lower than that reported for hexagonal GaN heterostructures.⁷⁹ Furthermore, the energy distribution of D_{it} exhibits several distinct features. Most notably, D_{it} increases with both temperature and energy depth. These trends arise from thermally activated processes: At elevated temperatures, deeper lying interface trap states become thermally accessible, resulting in an increase in the number of active trap states as well as an apparent shift of the effective energy distribution toward lower energies with respect to the conduction band edge (see Fig. 8).

Finally, D_{it} shows a clear dependence on the GaN layer thickness. As shown in Fig. 9, the average interface trap state density, whose standard deviation varies between 7% and 20%, decreases with increasing GaN layer thickness. This trend indicates an improvement in the structural quality of the properties of the existing interfaces with extended growth time. This observation is consistent with our Mott-Schottky analysis (Fig. 4), which confirms an improved structural quality by the reduced impurity concentration at extended growth times.

CONCLUSION

We characterized the electrical properties of the cubic GaN/AlN heterostructures grown on 3C-SiC/Si templates using impedance spectroscopy, revealing transport processes that are absent in the 3C-SiC/Si template. In particular, the cubic GaN/AlN heterostructures exhibit an additional feature under forward-bias, which we attributed to carrier accumulation at the c-AlN/3C-SiC interface in agreement with qualitative band structure simulations performed using nextnano++. Mott-Schottky analysis shows a systematic reduction of the impurity density with increasing c-GaN film, indicating a continuous improvement in crystal quality with extended growth time. This trend in crystal quality is independently corroborated by the observed decrease in interface trap density at the c-AlN/3C-SiC interfaces, highlighting the strong correlation between structural and electronic quality of the heterointerface. These results demonstrate that the structural properties of cubic nitride thin films can be tuned through growth optimization. This tunability is particularly of interest, as advancing the crystal quality of cubic nitrides represents a key step toward unlocking their full potential as a versatile platform for next-generation electronic and optoelectronic devices with enhanced functionalities.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for estimates of the resistances, capacitances, and time constants of the individual sample layers. Furthermore, it displays the impedance spectra of all measured samples, along with their determined fit parameters (resistance, capacitance, and time constant). Additionally, the parameters used for the band simulations are also given. The simulation procedure is described, and the resulting band diagrams are shown for all interfaces present in the sample.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Hannes Hergert: Conceptualization (supporting); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Project administration (supporting); Validation (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (supporting). **Silas A. Jentsch:** Data curation (supporting); Resources (supporting); Writing – review & editing (supporting). **Mario F. Zscherp:** Data curation (supporting); Resources (supporting); Writing – review & editing (supporting). **Jörg Schörmann:** Resources (supporting); Writing – review & editing (supporting). **Peter J. Klar:** Funding acquisition (equal); Resources (equal); Software (equal); Supervision (supporting); Writing – review & editing (supporting). **Sangam Chatterjee:** Funding acquisition (equal); Project administration (equal); Software (equal). **Matthias T. Elm:** Conceptualization (lead); Formal analysis (supporting); Funding acquisition (equal); Investigation (supporting); Methodology (supporting); Project administration (lead); Resources (lead); Software (equal); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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